AS · Unit 1 · WCH11 | Structure, Bonding and Introduction to Organic Chemistry

Atomic structure and the periodic table

Everything about how an element behaves comes from the arrangement of electrons around its nucleus. This topic starts with the particles inside an atom and the mass spectrometer that weighs them, then uses ionisation energies as the experimental evidence for shells, sub-shells and orbitals. It finishes by explaining the repeating patterns, or periodicity, across the periodic table.

SPEC 2.1 The structure of an atom

WHAT THIS SPEC POINT ASKS

Know that an atom is made of protons, neutrons and electrons, and where each is found.

- An atom has a tiny, dense, positively charged **nucleus** at its centre, containing **protons** and **neutrons** (together called **nucleons**).
- **Electrons** occupy the space around the nucleus, arranged in **shells** (energy levels), which are split into **sub-shells** and **orbitals** (2.9).
- Almost all the **mass** is in the nucleus; almost all the **volume** is the space where the electrons are. The nucleus is roughly 10^{-5} times the diameter of the atom.
- An atom is **neutral** because it has equal numbers of protons and electrons.

EXAM TIP

Use the right word for where electrons are: "shell" (energy level, n = 1, 2, 3…), "sub-shell" (s, p, d), "orbital" (a region holding up to 2 electrons). Examiners don't treat them as the same.

MARK-SCHEME WORDING

- "Protons and neutrons in the nucleus; electrons in shells / energy levels around it."
- Electrons are **not** in the nucleus: the nucleon count is protons + neutrons only.

Often asked as: part of a 2.4 counting question, or an MCQ on where most of the mass is.

COMMON MISTAKE

Counting electrons as nucleons. Describing electrons as moving in fixed circular "orbits" when the question asks about orbitals.

SPEC 2.2 Relative mass and charge of protons, neutrons and electrons

WHAT THIS SPEC POINT ASKS

Know the relative masses and relative charges of the three sub-atomic particles.

Particle	Where	Relative mass	Relative charge
proton	nucleus	1	+1
neutron	nucleus	1	0
electron	shells around the nucleus	1/1836 (about 0.0005, "negligible")	−1

- Protons and neutrons have almost the same mass; an electron is about 1/1840 of a proton.
- Use **relative** values in answers. (Actual values, such as 1.60×10^{-19} C for the charge, aren't needed.)

MARK-SCHEME WORDING

- Electron mass "1/1836" or "1/1840" or "negligible"; charge "−1" (a minus sign is needed). Proton +1, neutron 0.

Often asked as: complete the table; MCQ on which particle is deflected most (lightest) or not at all (neutral).

COMMON MISTAKE

Giving the electron's relative mass as 0, or its charge without the minus sign.

SPEC 2.3 Atomic (proton) number and mass number

WHAT THIS SPEC POINT ASKS

Know what atomic (proton) number and mass number mean.

- **Atomic number (proton number), Z**: the number of **protons** in the nucleus. It defines the element: every atom of carbon has 6 protons.
- **Mass number (nucleon number), A**: the total number of **protons + neutrons** in the nucleus.

- Nuclide notation: the mass number goes top left and the atomic number bottom left, e.g. $^{23}_{11}Na$. Often only the mass number is written: ^{23}Na, sodium-23.
- The Periodic Table lists elements in order of atomic number.

MARK-SCHEME WORDING

- Atomic number = "number of protons (in the nucleus of an atom)". Mass number = "number of protons **plus** neutrons".

Often asked as: part of 2.4 and 2.5 questions; MCQs comparing species.

COMMON MISTAKE

Saying atomic number is "the number of electrons": that's only true for a neutral atom. Writing the mass number when the question asks for the number of neutrons.

SPEC 2.4 Working out the particles in an atom or ion

WHAT THIS SPEC POINT ASKS

Use atomic number and mass number to find the number of protons, neutrons and electrons in an atom or ion.

- **Protons = Z**.
- **Neutrons = A − Z**.
- **Electrons = Z − (charge)**: a positive ion has **lost** electrons; a negative ion has **gained** them.

WORKED EXAMPLE

Worked example: counting particles

Species	Protons	Neutrons	Electrons
^{25}Mg	12	25 − 12 = 13	12
$^{25}Mg^{2+}$	12	13	12 − 2 = 10
$^{35}Cl^{-}$	17	35 − 17 = 18	17 + 1 = 18
$^{56}Fe^{3+}$	26	30	23
$^{18}O^{2-}$	8	10	10

Species with the **same number of electrons** are **isoelectronic**: Na^+, Mg^{2+}, F^- and O^{2-} all have 10 (3.7).

MARK-SCHEME WORDING

- Give the three **numbers**, labelled. A mass or atomic number in place of a particle count scores nothing.

Often asked as: complete a table for an atom and its ion; MCQ "which ion is not isoelectronic with the others?" (P^{3-}, Cl^- and Ca^{2+} have 18 electrons; Ti^{3+} has 22 − 3 = 19).

COMMON MISTAKE

Changing the number of protons when an ion forms: only **electrons** change. Adding electrons for a positive ion. Using the wrong isotope's mass number.

SPEC 2.5 Isotopes

WHAT THIS SPEC POINT ASKS

Understand what isotopes are.

- **Isotopes** are atoms of the **same element** with the **same number of protons** but **different numbers of neutrons** (so different mass numbers).
- Example: chlorine-35 (17 p, 18 n) and chlorine-37 (17 p, 20 n). Hydrogen has three: ^1H (no neutrons), ^2H (deuterium, 1 neutron) and ^3H (tritium, 2 neutrons).

Same chemistry, different physical properties

- Isotopes of an element have the **same electronic configuration**, so they react in the same way. Chemical reactions involve electrons, not neutrons.
- They have **different masses**, so physical properties that depend on mass differ slightly: density, rate of diffusion, melting and boiling points of their compounds, and how far they are **deflected** in a mass spectrometer (2.6).

MARK-SCHEME WORDING

- Isotope: "atoms of the same element with the same number of protons but different numbers of neutrons". The word **number** must be there. Electrons can be ignored if they equal the protons.
- "Same element" on its own does not replace "same number of protons".
- Same chemical reactions because they have the "same **electronic configuration**". "Same number of electrons" or "same outer electrons" did not score.

- In a mass spectrometer: different mass → different deflection (both halves needed).

Often asked as: define isotope; give a similarity and a difference between two isotopes; explain why isotopes react identically.

COMMON MISTAKE

Writing "same protons, different neutrons" without "number of". Saying isotopes have "different numbers of electrons" or "different chemical properties".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Atoms of **different** elements with the **same mass number** (e.g. germanium-76 and selenium-76): same number of nucleons but different numbers of protons, so they are different elements, not isotopes (WCH11 Jan 2025).

SPEC 2.6 The mass spectrometer and interpreting mass spectra

WHAT THIS SPEC POINT ASKS

Know the basic principles of a mass spectrometer and use mass spectra to: (i) find the isotopic composition of an element; (ii) calculate A_r from isotope abundances and vice versa; (iii) find the M_r of a molecule and identify it; (iv) understand that some ions have a 2+ charge.

How it works

Stage	What happens	Why
Vaporisation	the sample is turned into a gas	so particles can move separately
Ionisation	atoms or molecules are **bombarded with high-energy electrons** from an electron gun, knocking an electron out: $X(g) + e^- \rightarrow X^+(g) + 2e^-$	only **charged** particles can be accelerated and deflected
Acceleration	positive ions are accelerated by an **electric field** (negatively charged plates)	to give a fast beam

Stage	What happens	Why
Deflection	a **magnetic field** bends the ions' path; **lighter** ions (lower m/z) and more highly charged ions are deflected **more**	separates ions by mass-to-charge ratio
Detection	ions hit the detector and make a current; the bigger the current, the more ions	measures relative abundance

- The whole instrument is under a **vacuum**: otherwise ions would **collide with air molecules** and be deflected or lose their charge.
- Output: a **mass spectrum**, relative abundance (or intensity) against **m/z** (mass ÷ charge). For 1+ ions, m/z equals the mass.

(i) Isotopic composition

Each peak in the spectrum of an element is one isotope. The m/z gives its mass number; the height gives its relative abundance.

(ii) Relative atomic mass from a spectrum

> A_r = Σ (isotopic mass × abundance) ÷ Σ abundances

WORKED EXAMPLE

Worked example: A_r from percentages and from peak heights

(a) Magnesium: ^{24}Mg 78.99%, ^{25}Mg 10.00%, ^{26}Mg 11.01%.

A_r = (24 × 78.99 + 25 × 10.00 + 26 × 11.01) ÷ 100 = 2432.02 ÷ 100 = **24.32**

(b) Chlorine peaks: m/z 35, height 12.0; m/z 37, height 3.9.

A_r = (35 × 12.0 + 37 × 3.9) ÷ (12.0 + 3.9) = 564.3 ÷ 15.9 = **35.49**

Divide by the **total abundance** (15.9), not by the number of isotopes.

WORKED EXAMPLE

Worked example: working backwards

(a) Boron (A_r 10.8) contains only ^{10}B and ^{11}B. Let the % of ^{10}B be x.

10x + 11(100 − x) = 1080 → 1100 − x = 1080 → x = **20% ^{10}B**, 80% ^{11}B.

(b) Magnesium (A_r 24.3) is 79% ^{24}Mg; the other 21% is ^{25}Mg and ^{26}Mg. Let the fraction of ^{26}Mg be y.

24.3 = 24 × 0.79 + 25(0.21 − y) + 26y = 18.96 + 5.25 + y → y = 0.09

So **9% ^{26}Mg** and 12% ^{25}Mg.

(iii) M_r of a molecule

- When a molecule is ionised whole, it gives the **molecular ion, M^+**. The molecular ion peak is the peak with the **highest m/z** (ignoring the tiny M+1 peak from carbon-13), and its m/z = **M_r**.
- Molecular ions can also break up into **fragment ions**, which give peaks at lower m/z. The tallest peak (the **base peak**) is often a fragment, not the molecular ion.
- Example: ethanol, C_2H_5OH, has M^+ at m/z **46**; propanal (M_r 58) and butanal (M_r 72) are told apart by their molecular ion peaks.

(iv) 2+ ions

- If two electrons are knocked out, the ion is **2+** and appears at **half** its mass: m/z = mass ÷ 2.
- $^{24}Mg^{2+}$ appears at m/z **12**; $^{28}Si^{2+}$ at m/z **14**; the ^{35}Cl–$^{37}Cl^{2+}$ ion (mass 72) appears at m/z **36**.
- 2+ ions are deflected more than 1+ ions of the same mass.

Relative atomic mass from percentages

1. A sample of silicon contains 92.2% ^{28}Si, 4.7% ^{29}Si and 3.1% ^{30}Si. Calculate A_r of silicon.
2. Multiply each isotopic mass by its abundance and add: $(28\times 92.2)+(29\times 4.7)+(30\times 3.1)=2581.6+136.3+93.0=2810.9$
3. Divide by 100: $A_r=28.1$

   The answer must lie between the lightest and heaviest isotope, and close to the most abundant one. That is a quick check on your arithmetic.

Relative atomic mass from peak heights

1. The mass spectrum of boron has peaks at m/z = 10 and 11, with heights of 23 and 92. Calculate A_r.
2. $\sum(\text{mass}\times\text{height})=(10\times 23)+(11\times 92)=230+1012=1242$

3. The heights add up to $23 + 92 = 115$, not 100, so divide by 115: $A_r = 1242 \div 115 = 10.8$

Working back to the abundances

1. Chlorine has isotopes of mass 35 and 37 and A_r = 35.5. Find the percentage of each isotope.
2. Let the percentage of ^{35}Cl be x, so ^{37}Cl is $100 - x$: $\dfrac{35x + 37(100 - x)}{100} = 35.5$
3. $3700 - 2x = 3550$, so $x = 75$. Chlorine is 75% ^{35}Cl and 25% ^{37}Cl, a ratio of 3 : 1.

MARK-SCHEME WORDING

- Ionisation: "bombarding / firing (high-energy / high-speed) electrons" at the sample, or the equation $X + e^- \rightarrow X^+ + 2e^-$. "Bombarded with electrons" without "high energy" lost a mark.
- Acceleration: "electric field / potential difference / charged plates". A **magnetic** field for acceleration is not accepted.
- Vacuum: "air molecules would be present" and "ions would collide with them" (both points).
- Heavier ions are deflected **less**.
- Show the **charge** on every ion you name; A_r to the s.f. or d.p. asked.

Often asked as: label the stages; why a vacuum is needed; which isotope is deflected most; A_r from a spectrum or table; find an unknown abundance with algebra; identify the molecular ion; explain a peak at half the expected m/z.

COMMON MISTAKE

- Describing the ions formed instead of **how** ionisation happens. Saying a magnetic field accelerates the ions.
- Taking the base peak as the molecular ion.
- For a 2+ ion of Cl_2 at m/z 36, writing "$^{36}Cl^+$" (there is no such isotope in the sample).
- Rounding A_r too early, or dividing by the number of isotopes.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Fragment ion vs molecular ion, and the base peak as the most abundant ion (WCH11 Jan 2021).

- Identifying a molecule from its molecular ion peaks, e.g. m/z 78 and 80 for a chloropropane (WCH11 Jan 2024).
- Labelling each peak with the isotopes it contains (WCH11 Jan 2026).

SPEC 2.7 Predicting the mass spectra of diatomic molecules

WHAT THIS SPEC POINT ASKS

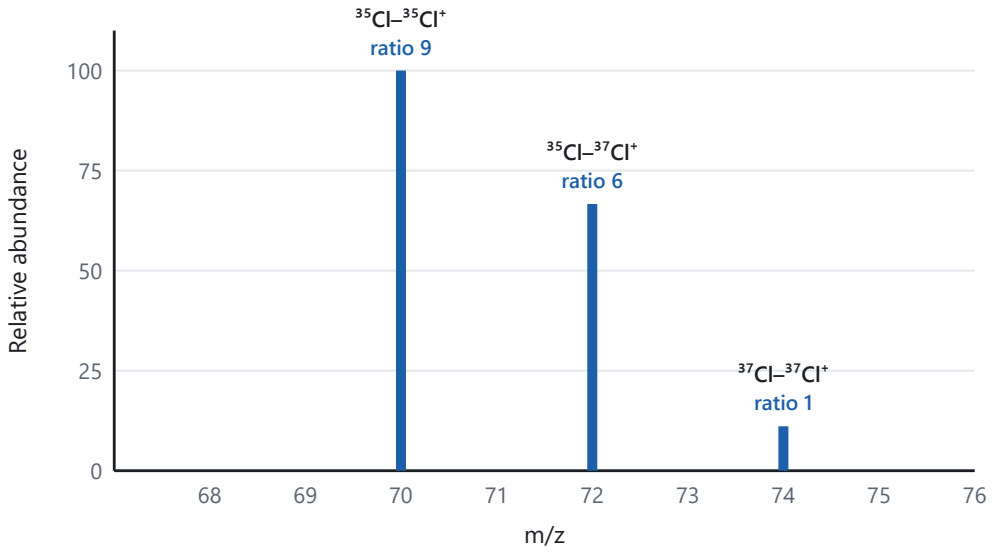
Predict mass spectra, including relative peak heights, for diatomic molecules such as Cl_2, given the isotopic abundances.

Chlorine: a probability grid

Chlorine is 75% ^{35}Cl and 25% ^{37}Cl, a ratio of **3 : 1**. Each Cl_2 molecule is two atoms picked at random:

	^{35}Cl (3)	^{37}Cl (1)
^{35}Cl (3)	$^{35}Cl^{35}Cl$, mass 70: 3 × 3 = **9**	$^{35}Cl^{37}Cl$, mass 72: 3 × 1 = 3
^{37}Cl (1)	$^{37}Cl^{35}Cl$, mass 72: 1 × 3 = 3	$^{37}Cl^{37}Cl$, mass 74: 1 × 1 = **1**

The mixed pair turns up twice, so its total is 3 + 3 = 6. Molecular ion peaks: **m/z 70 : 72 : 74 = 9 : 6 : 1**. (With percentages: $0.75^2 = 0.5625$, $2 \times 0.75 \times 0.25 = 0.375$, $0.25^2 = 0.0625$, the same ratio.)



Molecular-ion region of the Cl_2 spectrum. A real spectrum also has Cl^+ peaks at m/z 35 and 37 (3 : 1) from molecules that split.

Bromine

^{79}Br and ^{81}Br are about **1 : 1**, so Br_2 gives three peaks at **m/z 158 : 160 : 162 = 1 : 2 : 1**, plus Br^+ peaks at 79 and 81.

EXAM TIP

Draw vertical lines at the exact m/z values, with heights in the right ratio, and label each with its isotopes and a **+** charge. For a molecule with more halogen atoms, expand $(3 + 1)^n$ for chlorine: three Cl atoms give 27 : 27 : 9 : 1.

Predicting the Cl_2 peak heights

1. The chance that an atom is ^{35}Cl is $\frac{3}{4}$, and that it is ^{37}Cl is $\frac{1}{4}$.
2. m/z 70 needs ^{35}Cl twice: $\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$
3. m/z 72 can happen two ways, 35 then 37 or 37 then 35: $2 \times \frac{3}{4} \times \frac{1}{4} = \frac{6}{16}$

   Forgetting the factor of 2 is the most common mistake here.
4. m/z 74 needs ^{37}Cl twice: $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$. The peaks at 70, 72 and 74 are in the ratio **9 : 6 : 1**.

   The spectrum also shows peaks at m/z 35 and 37, in the ratio 3 : 1, from single Cl^+ ions made when some molecules break apart. Bromine is about 50% ^{79}Br and 50% ^{81}Br, so Br_2 gives peaks at 158, 160 and 162 in the ratio 1 : 2 : 1.

MARK-SCHEME WORDING

- **M1** peaks at the correct m/z; **M2** (needs M1) in the correct ratio. Any heights are accepted if the ratio is right.
- Explaining the ratio: "3 × 3 : 2 × 3 × 1 : 1 × 1 = 9 : 6 : 1", or a flow diagram of the four combinations.
- Br_2: middle peak double the outer two. The isotopes are ^{79}Br and ^{81}Br (there is no ^{80}Br).

Often asked as: sketch the Cl_2 or Br_2 spectrum; explain the 9 : 6 : 1 ratio; predict peaks for ClF_3 or NCl_3.

COMMON MISTAKE

- Only two peaks (70 and 74, or 158 and 162): you've missed the mixed molecule.
- Repeating 3 : 1 without multiplying out.

- Extra peaks at the wrong m/z; charges left off.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- ClF_3^+: fluorine has one isotope (^{19}F), so just two peaks, m/z 92 and 94, in a 3 : 1 ratio (WCH11 Oct 2019).
- NCl_3: four molecular ion peaks, 27 : 27 : 9 : 1 (WCH11 Jan 2019).

SPEC 2.8 First, second and third ionisation energies

WHAT THIS SPEC POINT ASKS

Define the first, second and third ionisation energies, and know that all ionisation energies are endothermic.

Definitions

First ionisation energy: the energy needed to remove **one electron** from **each atom** in **one mole of gaseous atoms**, to form **one mole of gaseous 1+ ions**.

$$Na(g) \rightarrow Na^+(g) + e^- \quad \Delta H = +496 \text{ kJ mol}^{-1}$$

Second ionisation energy: the energy needed to remove one electron from each ion in one mole of **gaseous 1+ ions**, to form one mole of gaseous 2+ ions.

$$Na^+(g) \rightarrow Na^{2+}(g) + e^- \quad \Delta H = +4562 \text{ kJ mol}^{-1}$$

Third ionisation energy: the same, starting from one mole of gaseous 2+ ions.

$$Na^{2+}(g) \rightarrow Na^{3+}(g) + e^- \quad \Delta H = +6910 \text{ kJ mol}^{-1}$$

Always endothermic

- The electron is **attracted** to the positive nucleus, so energy must be **put in** to pull it away. ΔH is always positive.
- Each successive ionisation energy is **larger**, because the electron is removed from an increasingly **positive** ion: the same number of protons pulls on fewer electrons, which are held more tightly and are closer to the

nucleus.

EXAM TIP

Write equations for **one** electron removed, with **(g)** on both species. Start from the right species: the second ionisation of lithium starts from $Li^+(g)$, the third of barium from $Ba^{2+}(g)$.

MARK-SCHEME WORDING

- Equations: **M1** correct species, balanced; **M2** state symbols (g) (depends on M1). Ignore state symbols on the electron. ALLOW $M(g) - e^- \rightarrow M^+(g)$.
- **Do not award multiples**: the definition is per mole of atoms, one electron each.

Often asked as: equation for the first IE of Cl or S, the second IE of Li, the third IE of Ba, $Ca^{2+} \rightarrow Ca^{3+}$, $F^+ \rightarrow F^{2+}$; define first IE.

COMMON MISTAKE

- Leaving out (g); writing S(s) for sulfur.
- Starting from Cl_2 instead of Cl(g).
- Putting the electron on the left (that's electron affinity), or confusing IE with bombarding atoms in a mass spectrometer.
- Writing $Na(g) \rightarrow Na^{2+}(g) + 2e^-$ for the second IE: that is the sum of the first and second.

SPEC 2.9 Orbitals

WHAT THIS SPEC POINT ASKS

Know that an orbital is a region within an atom that can hold up to two electrons with opposite spins.

- An **orbital** is a **region within an atom** where there is a high probability of finding an electron. It can hold **up to two electrons**, and if it holds two they have **opposite spins**.
- Opposite spins (drawn ↑↓) reduce the repulsion between the two electrons enough for them to share the orbital.

Shells, sub-shells and orbitals

Shell (n)	Sub-shells	Orbitals	Max electrons
1	1s	1	2
2	2s, 2p	1 + 3	2 + 6 = 8
3	3s, 3p, 3d	1 + 3 + 5	2 + 6 + 10 = 18
4	4s, 4p, 4d, 4f	1 + 3 + 5 + 7	2 + 6 + 10 + 14 = 32

A **sub-shell** is a group of orbitals of the same type and energy in one shell (an s sub-shell has 1 orbital, p has 3, d has 5, f has 7).

MARK-SCHEME WORDING

- "A region (**within** an atom) that can hold (up to) two electrons" and "with **opposite spins**". ALLOW spin +½ and −½.

Often asked as: define an orbital; why the two arrows in a box point opposite ways; MCQ on the number of orbitals or electrons in a sub-shell.

COMMON MISTAKE

- "A region **around** the atom": say **within**.
- Describing an orbital as an orbit or a path, or listing s, p, d instead of defining it.
- Using shell, sub-shell and orbital as if they meant the same thing.

SPEC 2.10 What affects the size of an ionisation energy

WHAT THIS SPEC POINT ASKS

Understand how ionisation energy depends on the number of protons, electron shielding and the sub-shell the electron is removed from.

An ionisation energy measures how strongly the **nucleus attracts** the electron being removed. These factors set it:

Factor	Effect on IE	Why
Nuclear charge (number of protons)	more protons → **higher** IE	stronger attraction for the electron

Factor	Effect on IE	Why
Distance from the nucleus (which shell)	further out → **lower** IE	attraction falls quickly with distance
Shielding by inner electrons	more inner shells → **lower** IE	inner electrons repel the outer electron and reduce the nucleus's pull
Sub-shell the electron is in	a p electron is easier to remove than an s electron in the same shell	the p sub-shell is at a slightly **higher energy** (and partly shielded by the s electrons)
Spin-pair repulsion	an electron in a **paired** orbital is easier to remove	the two electrons in one orbital repel each other

KEY IDEA

The template for any IE comparison: (1) say which has the **higher** value; (2) compare the **number of protons**; (3) compare the **shell** and **shielding**; (4) mention the **sub-shell** or **pairing** if it is an exception; (5) conclude with **attraction** between the nucleus and the outer electron.

MARK-SCHEME WORDING

- Use "nuclear charge" or "number of protons", "shielding (by inner shells)", "distance / energy level / shell", "sub-shell".
- "Effective nuclear charge" is allowed by some mark schemes and discouraged by some examiner reports: the safer wording is **"nuclear charge"** plus **"shielding"** as two separate points.

Often asked as: inside 2.11 and 2.18 explanations (trend across a period, down a group, the exceptions).

COMMON MISTAKE

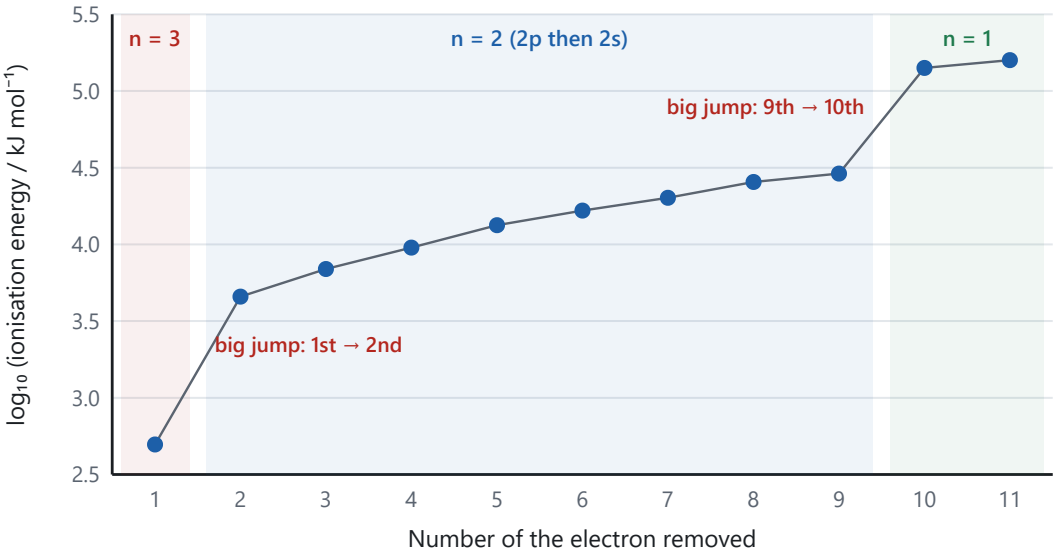
- Repeating one idea three times ("bigger radius, more shells, further away") and never mentioning nuclear charge.
- Saying shielding comes from electrons in the **same** sub-shell: shielding comes mainly from **inner** shells.

SPEC 2.11 Evidence for shells and sub-shells from ionisation energies

WHAT THIS SPEC POINT ASKS

Know how ideas about electronic structure grew from (i) successive ionisation energies, which show quantum shells and the group, and (ii) the first ionisation energies of successive elements, which show sub-shells.

(i) Successive ionisation energies: shells and the group



Sodium's 11 successive ionisation energies (log scale). Big jumps separate the shells: 1 electron in n = 3, 8 in n = 2, 2 in n = 1, so 2,8,1.

- Each IE is a bit bigger than the last (the ion gets more positive).
- A **big jump** means the next electron comes from a **new shell closer to the nucleus**: less shielded and much more strongly attracted.
- The **number of electrons removed before the first big jump = the number of outer-shell electrons = the group** (for s- and p-block elements).
- Smaller steps inside a shell show **sub-shells**: within n = 2 of sodium, the six 2p electrons (2nd–7th) come off before the two 2s electrons (8th and 9th).

WORKED EXAMPLE

Worked example: which group?

Successive IEs of an element (kJ mol^{-1}): 578, 1817, 2745, **11 577**, 14 842…

The big jump is between the **3rd and 4th**. So 3 electrons are in the outer shell: **Group 13** (this is aluminium).

Magnesium's 738, 1451, **7733** jumps between the 2nd and 3rd: Group 2.

Why use a log scale?

- The values range from hundreds to over 100 000 kJ mol^{-1}: on an ordinary scale the small ones would all squash onto the axis. Taking **$\log_{10}$** makes them all fit, and the jumps still show.
- To get a value back from a log, use the **antilog**: IE = 10^{x}. If $\log_{10}(IE) = 3.98$, IE = $10^{3.98}$ = 9550 kJ mol^{-1}.

(ii) First ionisation energies of successive elements: sub-shells

Across Period 2 (and 3) the first IE rises overall, but it **drops** in two places (see the graph in 2.17 and 2.18):

- **Be → B** (and Mg → Al): boron's outer electron is in **2p**, a higher-energy sub-shell than 2s, and is shielded by the 2s electrons. This dip shows that shells contain **sub-shells of different energy**.
- **N → O** (and P → S): oxygen's outer electron is the first to **pair up** in a 2p orbital; spin-pair repulsion makes it easier to remove. This dip shows that sub-shells are made of **orbitals** that hold two electrons each.

Finding the group from successive ionisation energies

1. The first five ionisation energies of an element, in kJ mol^{-1}, are 578, 1817, 2745, 11 577 and 14 842. Which group is it in?
2. Compare each value with the one before: ×3.1, ×1.5, **×4.2**, ×1.3. The biggest jump is between the 3rd and 4th ionisation energies.
3. So three electrons are relatively easy to remove and the fourth comes from an inner shell. The element has three outer electrons and is in **group 13** (it is aluminium).

Look for the ratio, not just the difference: later ionisation energies are always bigger in absolute terms.

MARK-SCHEME WORDING

- Group: the big jump shows "the electron is removed from a new / inner shell (closer to the nucleus)", and the number of electrons before it is the group number. Quote the actual electrons ("between the 3rd and

4th").

- General increase: "each electron is removed from a more positive ion" / "less repulsion between the remaining electrons".
- Log graph marks: axes labelled with units, e.g. "$\log_{10}$ (IE / kJ mol^{-1})"; points plotted; a scale using at least half the grid; the line does **not** go through the origin (there is no "electron 0").
- Why logs: "the range of values is too large" to plot on a normal scale.

Often asked as: identify the group or element from a set of successive IEs; plot $\log_{10}$ IE against electron number and read off a value by antilog; explain the jump and the general increase; why two successive IEs are close together (same sub-shell).

COMMON MISTAKE

- Linking the jump to "the group" rather than to a **new shell**.
- Not being able to do the antilog (10^x on the calculator).
- Misreading 4.06 as 4.60 on a log graph.
- Writing 2nd → 4th instead of the two electrons either side of the jump.

⚠ Beyond the spec · not part of the Edexcel IAL specification

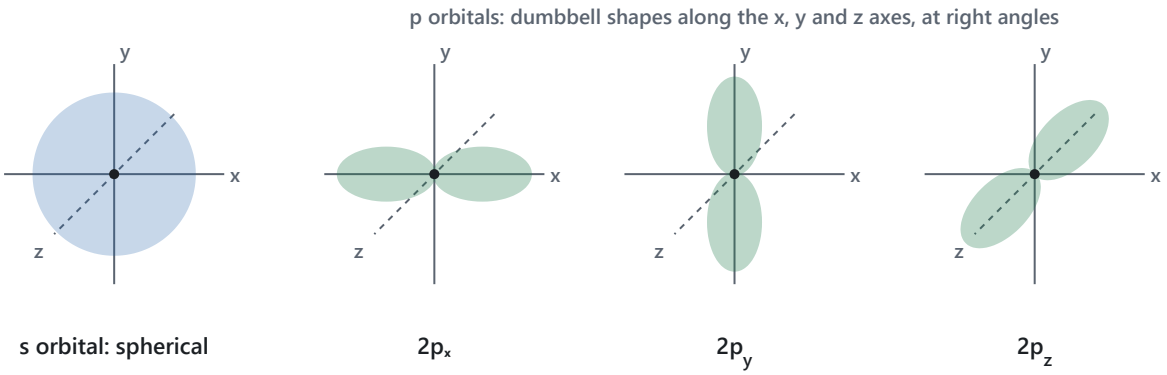
Examined in past papers even though the spec doesn't list it.

- Writing the shell structure (2,8,1) from the position of the jumps, not spd notation (WCH11 Oct 2020).
- Explaining a small rise inside a shell, e.g. between two electrons in the same sub-shell (WCH11 Jan 2022).

SPEC 2.12 Shapes of s and p orbitals

WHAT THIS SPEC POINT ASKS

Describe the shapes of s and p orbitals.



The nucleus is at the centre of each orbital, where the axes cross.

- **s orbitals** are **spherical**, centred on the nucleus. Each shell has one; it gets bigger as the shell number rises (2s is bigger than 1s).
- **p orbitals** are **dumbbell-shaped** (two lobes either side of the nucleus). There are **three** in each p sub-shell, at **right angles** to each other along the x, y and z axes (p_x, p_y, p_z). They have the same energy. The first p orbitals are in shell 2.
- You don't need the shapes of d orbitals.

MARK-SCHEME WORDING

- "Sphere / spherical". "Circle" or "round" alone is not awarded.
- p: "dumbbell / figure-of-eight", drawn along an axis with the nucleus where the lobes meet.

Often asked as: sketch an s and a p orbital; MCQ on shape and number of electrons in a p orbital.

COMMON MISTAKE

Drawing a p orbital as two separate blobs not touching at the nucleus. Saying a p orbital holds 6 electrons: the **sub-shell** holds 6; each **orbital** holds 2.

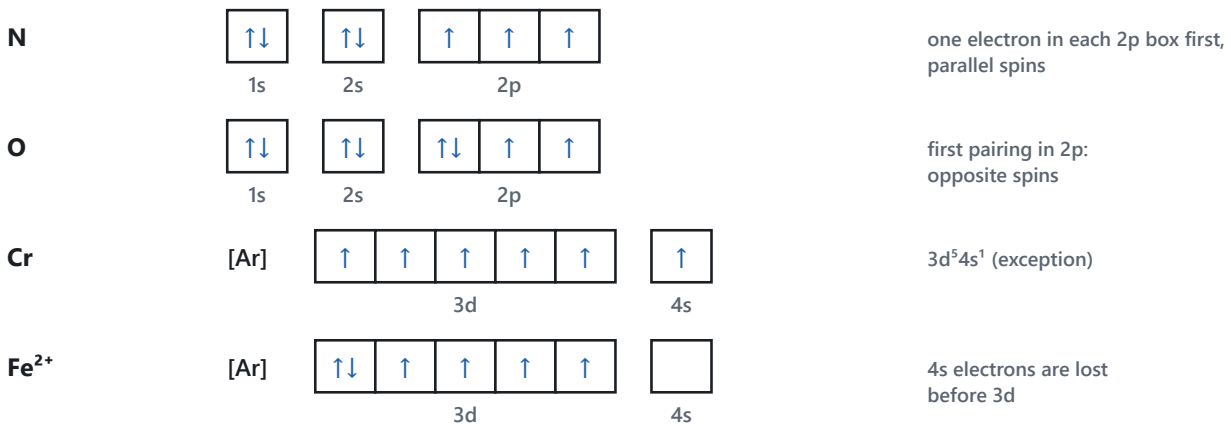
SPEC 2.13 How electrons fill orbitals

WHAT THIS SPEC POINT ASKS

Know that electrons in a sub-shell occupy orbitals singly before pairing up, and that paired electrons have opposite spins.

- Electrons fill the **lowest-energy** sub-shell available first.

- Within a sub-shell (e.g. the three 2p orbitals), electrons go **one into each orbital first**, with **parallel spins**, before any pair up. Being in separate orbitals keeps them further apart, so there is less repulsion. (This is Hund's rule.)
- When electrons pair up in an orbital, they have **opposite spins**. No orbital holds more than two.



Electrons-in-boxes: one box per orbital, one arrow per electron. Label every sub-shell.

MARK-SCHEME WORDING

- Boxes: **M1** sub-shells correctly labelled; **M2** boxes correctly filled. ALLOW half-headed arrows, labels $2p_x/2p_y/2p_z$, and any p box for the pair.
- DO NOT AWARD vertical lines without arrowheads, or pairs in more than one p box when the sub-shell isn't full.

Often asked as: draw sulfur or phosphorus in boxes; MCQ on the number of unpaired electrons (Cr has six).

COMMON MISTAKE

Pairing electrons in a p orbital while another p orbital in the same sub-shell is still empty (e.g. drawing S with two paired 3p boxes). Both arrows in a pair pointing the same way.

SPEC 2.14 Electronic configurations from hydrogen to krypton

WHAT THIS SPEC POINT ASKS

Predict the electronic configurations of atoms and ions from H to Kr, using s, p, d notation and electrons-in-boxes.

Filling order

1s → 2s → 2p → 3s → 3p → 4s → 3d → 4p

- **4s fills before 3d** because the 4s sub-shell is slightly lower in energy when it is empty.
- Shorthand: write the previous noble gas in square brackets, e.g. K = [Ar]$4s^1$.

Element	Configuration
N (7)	$1s^2\ 2s^2\ 2p^3$
Na (11)	$1s^2\ 2s^2\ 2p^6\ 3s^1$
Cl (17)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
Ca (20)	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$
Ti (22)	[Ar] $3d^2\ 4s^2$
Cr (24)	**[Ar] $3d^5\ 4s^1$** (not $3d^4 4s^2$)
Fe (26)	[Ar] $3d^6\ 4s^2$
Cu (29)	**[Ar] $3d^{10}\ 4s^1$** (not $3d^9 4s^2$)
Zn (30)	[Ar] $3d^{10}\ 4s^2$
Br (35)	[Ar] $3d^{10}\ 4s^2\ 4p^5$
Kr (36)	[Ar] $3d^{10}\ 4s^2\ 4p^6$

- **Chromium and copper** are the two exceptions: one 4s electron moves into 3d, giving a **half-filled** ($3d^5$) or **full** ($3d^{10}$) d sub-shell, which is a lower-energy (more stable) arrangement.
- Either order is accepted when writing: [Ar]$3d^6 4s^2$ or [Ar]$4s^2 3d^6$.

Ions

- **Negative ions**: add electrons to the next available orbital. Cl^- = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (same as Ar).
- **Positive ions of s- and p-block elements**: remove the outer electrons. Na^+ = $1s^2\ 2s^2\ 2p^6$.
- **Transition metal ions**: the **4s electrons are lost first**, before any 3d electrons.

Ion	Configuration
Fe^{2+}	[Ar] $3d^6$
Fe^{3+}	[Ar] $3d^5$
Cu^{2+}	[Ar] $3d^9$ (one 4s and one 3d electron removed)

Ion	Configuration
Ni^{2+}	[Ar] $3d^8$
Ti^{3+}	[Ar] $3d^1$
Zn^{2+}	[Ar] $3d^{10}$

MARK-SCHEME WORDING

- Cl: "$1s^2 2s^2 2p^6 3s^2 3p^5$"; ALLOW $2p_x^2 2p_y^2 2p_z^2$ for $2p^6$; [Ne] accepted for $1s^2 2s^2 2p^6$.
- Cu^{2+} = $[Ar]3d^9$: one electron from 4s and one from 3d, **not** both from 3d.

Often asked as: configuration of Cl, Ca, Al, P or S (in boxes); Cu and Cu^{2+}; Ni^{2+}; MCQ "which ion has this configuration?"

COMMON MISTAKE

- Writing 3d before 4s **in the filling**, or removing 3d electrons before 4s when forming an ion.
- Giving the GCSE form 2,8,8,2 when s, p, d notation is asked (or the reverse).
- Giving the atom's configuration when the ion was asked.
- Writing 4d for 3d; untidy superscripts.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Ions isoelectronic with argon (P^{3-}, S^{2-}, Cl^-, K^+, Ca^{2+}) all have the configuration $1s^2 2s^2 2p^6 3s^2 3p^6$. Ti^{3+} is the odd one out: $[Ar]3d^1$, 19 electrons (WCH11 Jan 2026).

SPEC 2.15 Electronic configuration determines chemical properties

WHAT THIS SPEC POINT ASKS

Understand that an element's electronic configuration determines its chemical properties.

- Chemical reactions involve the **outer electrons** (they are lost, gained or shared). So elements with the **same outer-shell configuration** react in similar ways: that's why elements in a **group** behave alike.

Group	Outer configuration	Typical behaviour
1	ns^1	lose 1 electron → M^+ (Na^+, K^+)
2	ns^2	lose 2 electrons → M^{2+}
13	$ns^2\ np^1$	Al forms Al^{3+}; three outer electrons
16	$ns^2\ np^4$	gain 2 electrons → X^{2-} (O^{2-}, S^{2-})
17	$ns^2\ np^5$	gain 1 electron → X^-; or share one electron in a covalent bond
18	$ns^2\ np^6$ (He $1s^2$)	full outer sub-shells, very high IE: very unreactive

- The **number of shells** changes down a group, so reactivity changes (e.g. Group 1 metals become more reactive because the outer electron is further away and easier to lose).
- Isotopes have identical chemistry because their configurations are identical (2.5).

MARK-SCHEME WORDING

- Same group → "same number of electrons in the outer shell / same outer electronic configuration". Different reactivity → "different number of shells".

Often asked as: MCQ predicting a formula (sulfur gains 2 electrons, so it forms K_2S with potassium); compare Group 13 elements (three outer electrons, different numbers of shells).

COMMON MISTAKE

Saying elements in a period have similar chemistry: it's elements in a **group**.

SPEC 2.16 Blocks of the Periodic Table and sub-shell capacities

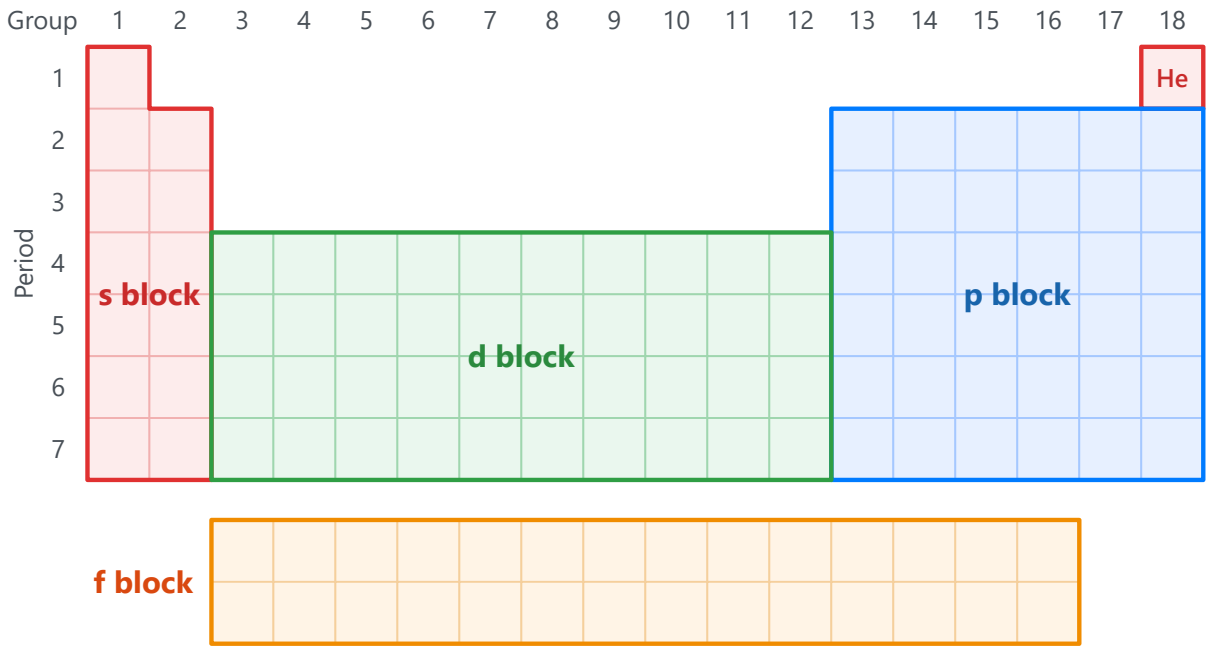
WHAT THIS SPEC POINT ASKS

Know that the Periodic Table is divided into s, p and d blocks, and know how many electrons s, p and d sub-shells hold in the first four shells.

- An element's **block** is the sub-shell its **highest-energy electron** goes into.

Block	Groups	Sub-shell being filled
s	1 and 2 (and He)	s (holds 2)
p	13 to 18	p (holds 6)
d	the transition elements, Sc to Zn in Period 4	d (holds 10)
f	lanthanides and actinides	f (holds 14)

- Capacities: **s 2, p 6, d 10, f 14**. Shell totals: n = 1 holds 2, n = 2 holds 8, n = 3 holds 18, n = 4 holds 32.
- Helium is placed in Group 18 because it has a full outer shell, but its configuration ($1s^2$) makes it an s-block element.



The blocks of the periodic table. The block tells you which type of sub-shell the outermost electrons are filling. Helium ($1s^2$) sits in group 18 but is shaded as an s-block element.

MARK-SCHEME WORDING

- "s holds 2, p holds 6, d holds 10"; order of filling 4s before 3d.

Often asked as: MCQs on block, number of sub-shells, or electrons in a shell; state the size and order of sub-shells.

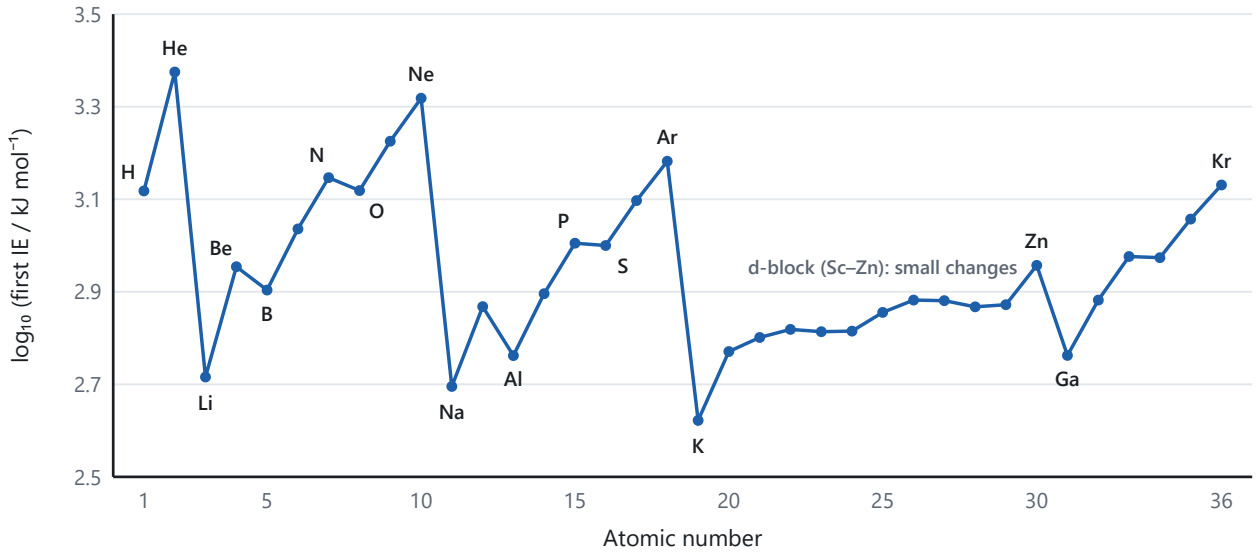
COMMON MISTAKE

Saying the third shell holds 8 electrons: it **can** hold 18 (3s, 3p and 3d). The "2,8,8" pattern only describes the order of filling up to calcium.

SPEC 2.17 Graphs of ionisation energy and periodic properties

WHAT THIS SPEC POINT ASKS

Plot data for elements 1 to 36 as graphs (including $\log_{10}$ of first ionisation energies) and use them to explain what a "periodic property" is.



$\log_{10}$ (first ionisation energy) for H to Kr. Peaks at the noble gases, troughs at the Group 1 metals, small dips at Groups 13 and 16.

What the graph shows

- **Peaks** at the noble gases (He, Ne, Ar, Kr): a full outer shell, so the electron is very hard to remove.
- **Troughs** at the Group 1 metals (Li, Na, K): one electron in a new shell, well shielded.
- The same **pattern repeats** across each period: a general rise, with dips at Group 13 (start of the p sub-shell) and Group 16 (first paired p electron).
- Across the d block (Sc to Zn) the first IE hardly changes: the electrons removed are 4s electrons, and the extra protons are largely shielded by the new 3d electrons.

Periodic property

A **periodic property** is one that shows a **regular repeating pattern** when plotted against **atomic number**, with the same trend repeated in each **period**. First ionisation energy, atomic radius and melting temperature are all periodic properties.

EXAM TIP

When plotting: choose a scale that uses at least half the grid on both axes, label axes with quantity and unit (e.g. "$\log_{10}$ (IE / kJ mol^{-1})"), plot every point within half a square, and join the points with straight lines or a smooth line. Don't extend the line to the origin.

MARK-SCHEME WORDING

- "Periodicity / periodic property": a "repeating pattern" or "trend repeated across each period".
- Graph: axes labelled, sensible linear scale, all points plotted.

Often asked as: plot first IEs or atomic radii against atomic number; describe the pattern; read a value off the graph.

COMMON MISTAKE

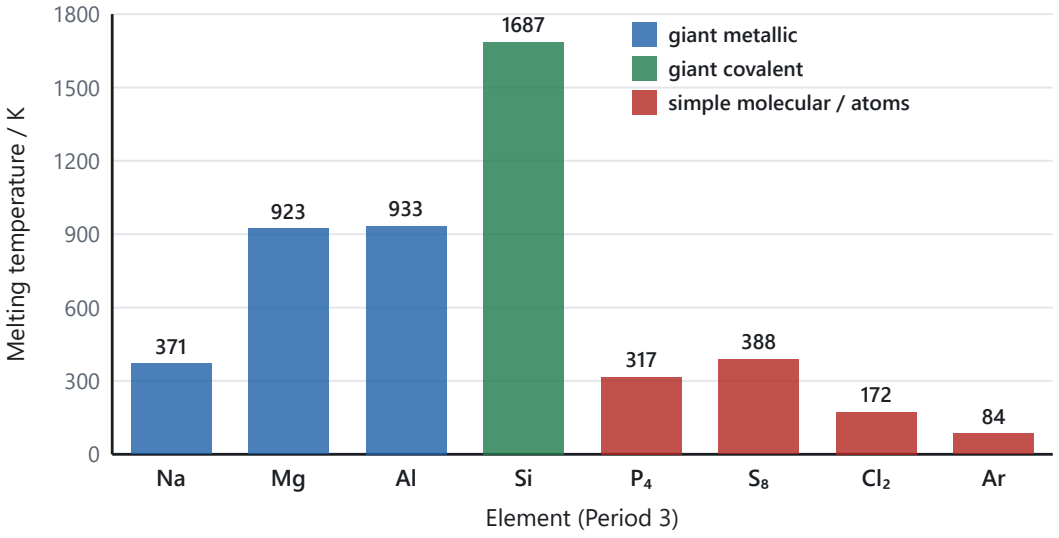
Describing a periodic property as "the property increases across the period" without saying the pattern **repeats**.

SPEC 2.18 Trends across Periods 2 and 3 and down a group

WHAT THIS SPEC POINT ASKS

Explain: (i) the trends in melting and boiling temperatures across Periods 2 and 3, in terms of structure and bonding; (ii) the general increase and the specific dips in ionisation energy across Periods 2 and 3; (iii) the decrease in first ionisation energy down a group.

(i) Melting and boiling temperatures

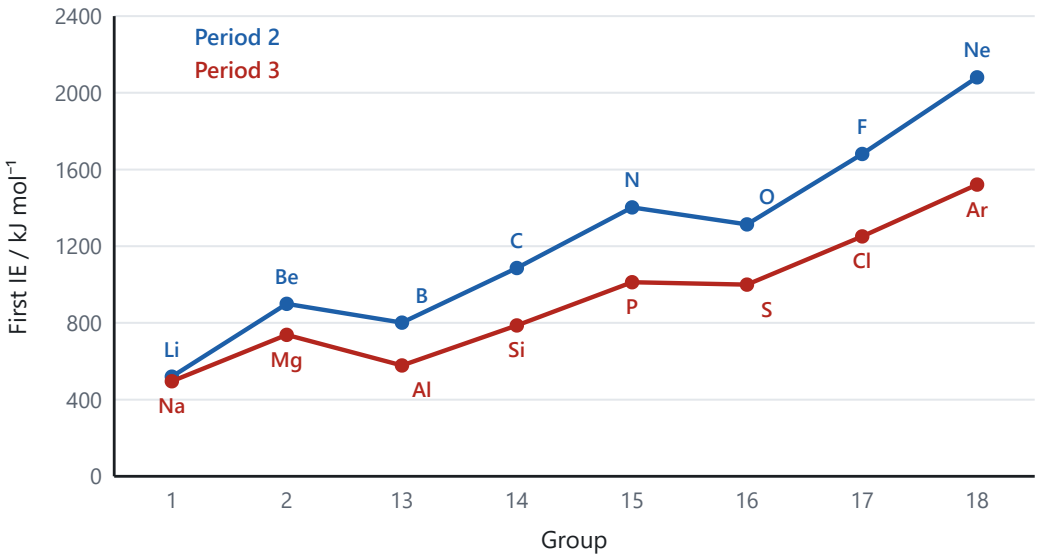


Melting temperatures across Period 3, coloured by structure.

Element(s)	Structure	What must be overcome on melting	Result
Na, Mg, Al	giant **metallic** lattice	strong attraction between metal ions and delocalised electrons	high; rises Na → Al
Si	giant **covalent** lattice	**many strong covalent bonds**	highest in the period
P_4, S_8, Cl_2	**simple molecular**	**weak London forces between molecules** (covalent bonds inside molecules don't break)	low
Ar	single **atoms**	very weak London forces between atoms	lowest

- **Na → Mg → Al**: the ions get **more highly charged** (Na^+, Mg^{2+}, Al^{3+}) and **smaller**, and each atom gives **more delocalised electrons** (1, 2, 3). So the metallic bonding gets stronger. (Boiling temperatures show this most clearly: Na 1156 K, Mg 1363 K, Al 2792 K.)
- **S_8 > P_4 > Cl_2 > Ar**: bigger molecules with **more electrons** have stronger London forces. S_8 (128 electrons) beats P_4 (60), Cl_2 (34) and Ar (18).
- **Period 2** follows the same pattern: Li (454 K) and Be (1560 K) are metallic, with Be stronger (Be^{2+}, two delocalised electrons per atom); **B** (2349 K) and **C** (about 3800 K, graphite/diamond) are giant covalent; N_2 (63 K), O_2 (54 K), F_2 (53 K) are simple molecules and Ne (25 K) is atoms: all very low.

(ii) Ionisation energy across Periods 2 and 3



First ionisation energies across Periods 2 and 3. Same shape in both; Period 3 values are lower (one more shell).

General increase across a period

- The **nuclear charge increases** (one more proton each time).
- The extra electrons go into the **same shell**, so **shielding stays about the same** and the atoms get smaller.
- So the outer electron is **attracted more strongly** and the IE rises.

Dip from Group 2 to Group 13 (Be → B, Mg → Al)

- B: $1s^2\ 2s^2\ \mathbf{2p^1}$. Its outer electron is in the **2p** sub-shell, which is **higher in energy** than 2s (and is partly shielded by the 2s electrons). It is easier to remove, despite the extra proton.

Dip from Group 15 to Group 16 (N → O, P → S)

- N: 2p has three **unpaired** electrons, one in each orbital. O: $2p^4$, so one 2p orbital now holds a **pair**.
- The **repulsion between the paired electrons** makes one of them easier to remove, so O's first IE is lower than N's.

(iii) Down a group

The first IE **decreases** down a group (e.g. Li 520 > Na 496 > K 419 kJ mol^{-1}; Cl 1251 > Br 1140 kJ mol^{-1}):

- The outer electron is in a **shell further from the nucleus** (larger atomic radius).
- There are **more inner shells**, so **more shielding**.
- The nuclear charge does increase, but this is outweighed by the extra distance and shielding.

- So there is **less attraction** between the nucleus and the outer electron.

MARK-SCHEME WORDING

- Melting: name the **structure and bonding** of each element, say what is broken or overcome, then compare the energy needed. Si: "giant covalent (lattice), strong covalent bonds throughout, a lot of energy to break". P_4: "simple molecular; weak intermolecular (London) forces; little energy to overcome". Giant covalent answers must not mention ions, metallic bonds or intermolecular forces.
- Across a period (2–3 marks): "more protons / nuclear charge increases"; "electron added to the same shell / similar shielding"; sometimes also "removed from the same sub-shell". IGNORE atomic radius in that mark scheme.
- N → O: O "loses a paired electron (from a 2p orbital)"; N "loses an electron from a singly occupied orbital"; "repulsion between paired electrons". The reverse argument is allowed.
- Mg → Al: Al's outer electron is in **3p**, Mg's in **3s**; 3p is "further from the nucleus / higher energy / shielded by the 3s electrons".
- Down a group: state **which is higher** first, then three of: further from the nucleus / more shells; more shielding; less attraction (despite more protons).

Often asked as: explain the trend across Period 3; explain one dip (often N/O or Mg/Al); compare two elements in a group; explain the melting temperatures of Si and P, or spot a wrongly plotted point on a melting-temperature graph.

COMMON MISTAKE

- Saying **covalent bonds break** when phosphorus or sulfur melts, or calling silicon "simple molecular".
- Writing "weak covalent bonds" or "strong structures" instead of naming the forces.
- Saying "a half-full p **orbital** is stable": it's a half-full **sub-shell** (one electron in each orbital), and you must say **which** electrons repel.
- Explaining a dip with the general trend, or stating a trend without explaining it.
- Answers with no comparator ("it is higher") or with three versions of the same distance point.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Estimating a missing melting temperature from the pattern: N_2 is a gas at room temperature, so its melting temperature is far below 298 K (WCH11 Oct 2024).
- Boron has a giant covalent structure and a very high melting temperature (WCH11 Oct 2024).
- Spotting impossible values on a plotted graph, e.g. argon at 500 K (WCH11 Jan 2020).

Quick checks

QUICK CHECK

How many protons, neutrons and electrons are in the ion $^{34}_{16}\mathrm{S}^{2-}$?

- 16, 18, 16
- 16, 18, 14
- 16, 18, 18
- 18, 16, 18

Z = 16 gives 16 protons, and $34 - 16 = 18$ neutrons. The 2− charge means two electrons have been gained: $16 + 2 = 18$ electrons. The first option forgets the charge; 14 electrons would be a 2+ ion.

QUICK CHECK

An element has two isotopes, of mass 63 (69.2%) and 65 (30.8%). What is its relative atomic mass?

- 64.0
- 63.6
- 63.0
- 64.4

$A_\mathrm{r} = (63 \times 69.2 + 65 \times 30.8)/100 = 63.6$. The simple average, 64.0, ignores the fact that the lighter isotope is more than twice as abundant.

QUICK CHECK

Chlorine is 75% ^{35}Cl and 25% ^{37}Cl. What are the relative heights of the Cl_2^+ peaks at m/z 70, 72 and 74?

- 3 : 1 : 0
- 9 : 3 : 1
- 1 : 2 : 1
- 9 : 6 : 1

$\frac{3}{4}\times\frac{3}{4}=\frac{9}{16}$, $2\times\frac{3}{4}\times\frac{1}{4}=\frac{6}{16}$ and $\frac{1}{4}\times\frac{1}{4}=\frac{1}{16}$. The 72 peak can form in two ways, which is why it is 6, not 3. A 1 : 2 : 1 ratio belongs to Br_2, whose isotopes are equally abundant.

QUICK CHECK

The first four ionisation energies of an element are 738, 1451, 7733 and 10 543 kJ mol^{-1}. Which group is the element in?

- 2
- 3
- 4
- 1

The big jump comes between the 2nd and 3rd values (1451 to 7733, more than five times bigger). Two electrons come from the outer shell and the third from a full inner shell, so the element is in group 2 (it is magnesium).

QUICK CHECK

Why is the first ionisation energy of oxygen lower than that of nitrogen?

- Oxygen has fewer protons than nitrogen.
- Oxygen's outer electron is in a shell further from the nucleus.
- Oxygen's electron is removed from a 2p orbital that holds two electrons, which repel each other.
- Oxygen's outer electron is in the 2s sub-shell.

Nitrogen has one electron in each 2p orbital. Oxygen's fourth 2p electron has to pair up, and repulsion between the two electrons in that orbital makes one of them easier to remove. Both elements lose their electron from the second shell, and oxygen has more protons, not fewer.

QUICK CHECK

What is the electron configuration of a chromium atom (Z = 24)?

- [Ar] $3d^4\ 4s^2$
- [Ar] $3d^5\ 4s^1$
- [Ar] $3d^6$
- [Ar] $4s^2\ 4p^4$

Chromium is an exception: a half-full 3d sub-shell, with one electron in each of the five d orbitals and one in 4s, is more stable than $3d^4\ 4s^2$, the configuration the simple filling order predicts.

QUICK CHECK

Which period 3 element has the highest melting temperature?

- Aluminium, because it has the strongest metallic bonding
- Sulfur, because S_8 molecules have the most electrons
- Magnesium, because its ions have a 2+ charge
- Silicon, because melting breaks many strong covalent bonds in a giant lattice

Silicon has a giant covalent structure, so melting means breaking strong covalent bonds throughout the lattice. Sulfur is simple molecular: melting only overcomes London forces between S_8 molecules, which are much weaker than covalent bonds.

QUICK CHECK

Why does the first ionisation energy decrease down group 1, even though the nuclear charge increases?

- The outer electron is paired, so it is repelled more.
- The outer electron is further from the nucleus and is shielded by more inner shells.
- The outer electron moves into a p sub-shell.
- Larger atoms have fewer protons per neutron.

Each step down adds a shell. The outer s electron is further away and has more full shells between it and the nucleus, and these two effects outweigh the extra protons, so less energy is needed to remove it.